

Davies' Ratio Limit for Locally Perturbed Cylindrical Tubes via the Zero-Threshold Spectral Measure

Literature-implied answer packet for arXiv:1105.0842

August 17, 2026

1 Status and source question

Status: literature-implied positive answer (full for the two classes asked about). This packet records an agent-identified consequence of an existing cylindrical-threshold spectral theorem. It is not presented as a new theorem of the supporting authors.

Grillo, Kovařík, and Pinchover ask in Appendix C, Remark C.2, on PDF page 22 of the current arXiv PDF [1] whether Davies' conjecture holds

for operators L satisfying the assumptions of Theorem 3.19 and for the Laplacian on a twisted tube.

Here Theorem 3.19 concerns the nonnegative self-adjoint Friedrichs realization of a real uniformly elliptic operator

$$L = -\nabla \cdot (a \nabla) + V$$

on the straight tube $\Omega_0 = \omega \times \mathbb{R}$, assumed subcritical and equal to $-\Delta - E_1$ outside a compact longitudinal region. The twisted case is the shifted Dirichlet Laplacian $H = -\Delta_\Omega^D - E_1$, where the twisting is compactly supported. The source proves, for either class and each fixed interior point x ,

$$c_x t^{-3/2} \leq e^{-tH}(x, x) \leq C_x t^{-3/2} \quad (t \geq t_x). \quad (1)$$

The decisive later input is Christiansen–Datchev's black-box spectral theory for compactly supported perturbations of cylindrical ends [2]. Their Sections 2.2–2.5 cover multiple ends and Dirichlet waveguides; Lemmas 2.4, 2.5, and 2.6 give threshold regularity, the generalized-eigenfunction spectral measure, and the form of a threshold resonance. The supporting paper does not mention Davies' conjecture or identify this application.

2 Identification and result

Both source classes fit the Christiansen–Datchev framework with

$$H_Y = -\Delta_\omega^D - E_1 \geq 0$$

on each cylindrical end. The zero eigenvalue of H_Y has finite multiplicity q (in the two-ended tube, $q = 2$), and the next transverse threshold ν_1 is positive. Let $\Phi_1(\lambda), \dots, \Phi_q(\lambda)$ be the generalized eigenfunctions associated with the zero channels, normalized as in [2, Section 2.3].

Theorem 1. *For every operator $H = L$ in source Theorem 3.19, and for the shifted Dirichlet Laplacian $H = -\Delta_\Omega^D - E_1$ of the locally twisted tube, there are locally continuous functions (with the elliptic regularity supplied by the coefficients)*

$$\Psi_j = \partial_\lambda \Phi_j(0), \quad 1 \leq j \leq q,$$

such that, for every fixed pair of interior points x, y ,

$$e^{-tH}(x, y) = t^{-3/2}B(x, y) + O_{x,y}(t^{-2}), \quad B(x, y) = \frac{1}{8\sqrt{\pi}} \sum_{j=1}^q \Psi_j(x) \overline{\Psi_j(y)}. \quad (2)$$

Moreover $B(x, y) > 0$ for all x, y . Consequently, for arbitrary fixed reference points x_0, y_0 ,

$$\lim_{t \rightarrow \infty} \frac{e^{-tH}(x, y)}{e^{-tH}(x_0, y_0)} = \frac{B(x, y)}{B(x_0, y_0)} > 0. \quad (3)$$

The same ratio limit holds for the unshifted twisted-tube Dirichlet heat kernel. Thus Davies' conjecture has a positive answer in both cases singled out in Remark C.2.

Proof. We give the derivation because the implication is not stated in the supporting paper.

1. *Exclusion of zero point spectrum.* A zero eigenfunction would contribute a nondecaying squared-modulus term to the diagonal heat kernel at every point where it is nonzero. This contradicts the upper bound in (1). Hence zero is not an eigenvalue of H .

2. *Exclusion of a zero-threshold resonance.* Let $R(\lambda) = (H - \lambda^2)^{-1}$, with $R(\lambda)$ and $R(-\lambda)$ denoting the two physical boundary values for $\lambda > 0$. Below the first positive transverse threshold, Christiansen–Datchev Lemma 2.5 says, after compact localization,

$$\frac{1}{i}(R(\lambda) - R(-\lambda)) = \frac{1}{2\lambda} \sum_{j=1}^q \Phi_j(\lambda) \otimes \Phi_j(\lambda), \quad 0 < \lambda < \nu_1, \quad (4)$$

apart from the point-spectrum projection, which has already been excluded at zero. If some $\Phi_j(0)$ were nonzero, choose x with $\Phi_j(0, x) \neq 0$. Threshold continuity from their Lemma 2.4 and the positive diagonal spectral density in (4) would imply

$$e^{-tH}(x, x) \geq c \int_0^\varepsilon e^{-t\lambda^2} d\lambda \geq c' t^{-1/2}$$

for all sufficiently large t , contradicting (1). Therefore

$$\Phi_j(0) = 0 \quad (1 \leq j \leq q). \quad (5)$$

By Christiansen–Datchev Lemma 2.6 and the discussion following it, this is exactly the absence of a zero-threshold resonance, so the cutoff resolvent is regular at zero.

3. *The exact heat-kernel asymptotic.* The generalized eigenfunctions are meromorphic in the threshold coordinate and regular at the physical threshold; at zero that coordinate is simply λ . Local elliptic regularity lets this expansion be evaluated pointwise on compact subsets. Hence (5) gives

$$\Phi_j(\lambda) = \lambda \Psi_j + O(\lambda^2). \quad (6)$$

Choose $0 < \delta < \nu_1$ so small that the meromorphically continued cutoff resolvent has no pole for $|\lambda| < \delta$. Stone's formula and (4) give the low-energy kernel

$$K_{<\delta^2}(t; x, y) = \frac{1}{2\pi} \int_0^\delta e^{-t\lambda^2} \sum_{j=1}^q \Phi_j(\lambda, x) \overline{\Phi_j(\lambda, y)} d\lambda. \quad (7)$$

Indeed, the factor 2λ from $d(\lambda^2)$ cancels the 2λ in (4). Substituting (6) and using

$$\int_0^\infty e^{-t\lambda^2} \lambda^2 d\lambda = \frac{\sqrt{\pi}}{4t^{3/2}}$$

yields (2) for the low-energy part, with an $O_{x,y}(t^{-2})$ remainder.

For completeness, the remaining spectrum is exponentially negligible pointwise. If E_H is the spectral resolution, insert the time-1/2 heat kernel on both sides of $E_H([\delta^2, \infty))e^{-(t-1)H}$. Since $\|e^{-H/2}\delta_x\|_2^2 = e^{-H}(x, x) < \infty$, Cauchy–Schwarz gives

$$|K_{\geq \delta^2}(t; x, y)| \leq e^{-(t-1)\delta^2} \sqrt{e^{-H}(x, x)e^{-H}(y, y)}.$$

This also absorbs any positive embedded eigenvalues away from zero.

4. *Strict positivity.* Taking $x = y$ in (2) and using the lower half of (1) gives $B(x, x) > 0$ at every interior point. Fixed-point parabolic Harnack comparison gives, for each fixed x, y , a positive lower comparison between $e^{-tH}(x, y)$ and a diagonal kernel at a boundedly shifted time. In the twisted-tube discussion the source states this pointwise equivalence explicitly, citing Davies’ Theorem 10 [1, PDF p. 2]. Passing to (2) gives $B(x, y) > 0$. Dividing the two asymptotics proves (3).

Finally, if $P = -\Delta_\Omega^D$, then $e^{-tP} = e^{-E_1 t} e^{-t(P-E_1)}$; the scalar exponential cancels from the ratio. $\square$

3 Scope, provenance, and review points

This settles exactly the self-adjoint uniformly elliptic class in source Theorem 3.19 and its locally twisted Dirichlet tube. It does not address arbitrary nonsymmetric parabolic operators or Davies’ conjecture on general noncompact manifolds.

The run’s registry, solution, attempt, and proof-gap indexes were searched for the source id and the core Davies/twisted-tube/ratio-limit terms. Bounded direct searches covered the exact source phrase, the source title, “Davies conjecture twisted tube,” and later heat/spectral asymptotics for cylindrical waveguides. No paper explicitly announcing this answer was found. The supporting paper supplies the decisive existing spectral identity and threshold regularity, but does not cite the source or discuss this ratio limit. The classification is therefore *literature-implied answer*, not *literature already answered*, and no originality claim is made.

The main human-review points are: (i) the black-box fit for a compactly supported coefficient perturbation after shifting by E_1 ; (ii) the localized Stone normalization in (7); (iii) the pointwise passage, justified by local elliptic regularity and the time-one smoothing estimate; and (iv) the Harnack step that upgrades a nonzero diagonal coefficient to strict positivity for every ordered pair.

References

- [1] G. Grillo, H. Kovařík, and Y. Pinchover, *Sharp two-sided heat kernel estimates of twisted tubes and applications*, arXiv:[1105.0842](#).
- [2] T. J. Christiansen and K. Datchev, *Wave asymptotics for waveguides and manifolds with infinite cylindrical ends*, arXiv:[1705.08972](#).